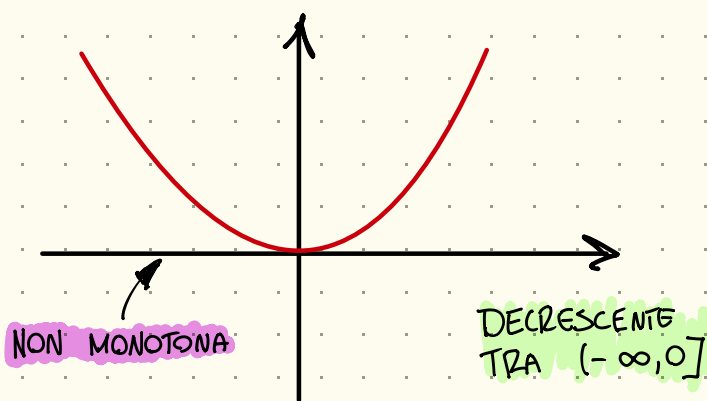
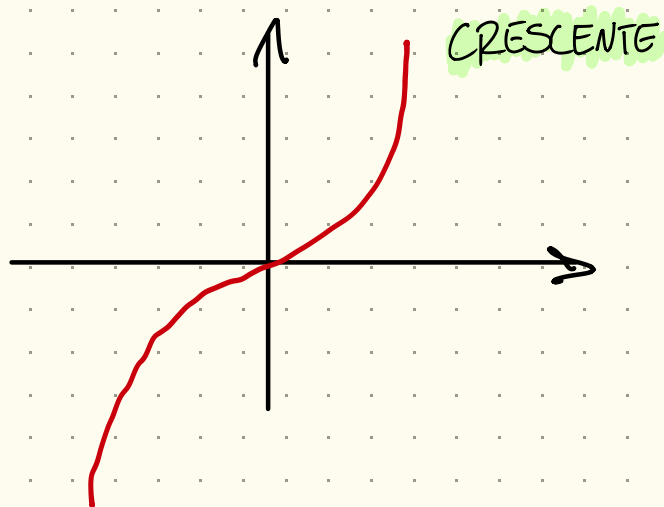
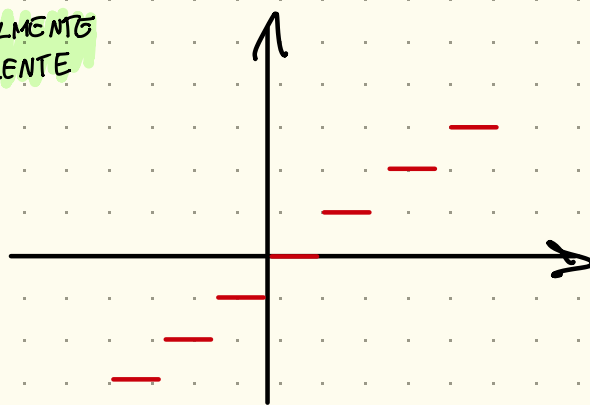


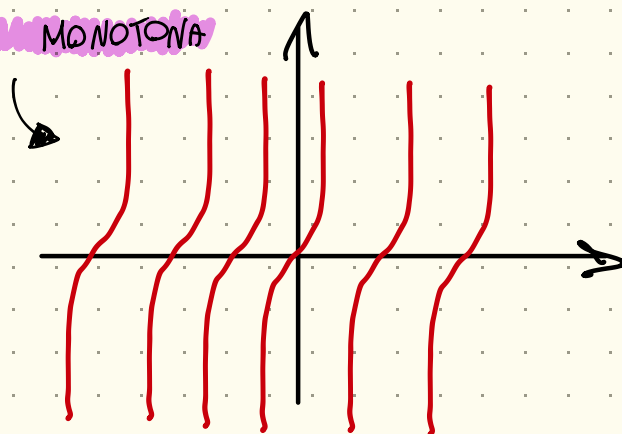
FUNZIONI MONOTONE



DEBOLMENTE
CRESCENTE



NON MONOTONA



- Def CRESCENTE

SE CON x_1 E $x_2 \Rightarrow y_1 < y_2$

• OPERAZIONI ALGEBRICHE

$$f: A \rightarrow \mathbb{R} \quad g: B \rightarrow \mathbb{R}$$

$$\bullet (f \pm g)(x) = f(x) \pm g(x)$$

$$\text{dom}(f \pm g) = A \cap B$$

- $(f \cdot g)(x) = f(x) \cdot g(x)$

$$\text{dom}(f \cdot g) = A \cap B$$

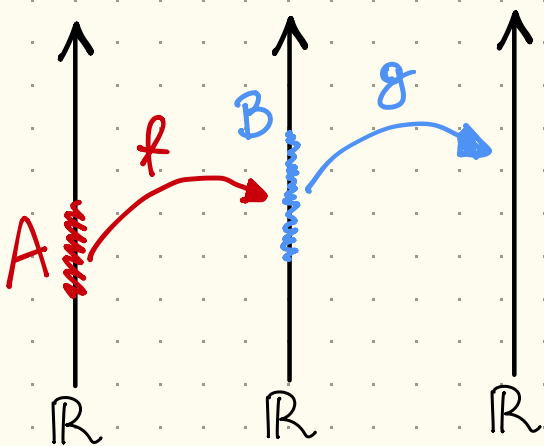
- $\frac{f}{g}(x) = \frac{f(x)}{g(x)}$

$$\text{dom} \frac{f}{g} = \{x \in A \cap B : g(x) \neq 0\}$$

- $\frac{1}{f}(x) = \frac{1}{f(x)}$ RECIPROCO

• FUNZIONE COMPOSTA

$$f: A \rightarrow \mathbb{R} \quad g: B \rightarrow \mathbb{R}$$



LA FUNZIONE COMPOSTA È:

$$(g \circ f)(x) = g(f(x))$$

$$\text{dom}(g \circ f) = \{x \in A : f(x) \in B\}$$

PRODOTTI

$$f \circ g = g \circ f$$

COMPOSIZIONE

$$g \circ f \neq f \circ g$$

es)

$$\bullet f(x) = x^2$$

$$\bullet g(x) = 1+x$$

$$\bullet \text{dom } f = \mathbb{R}, \text{ dom } g = \mathbb{R}$$

$$\begin{aligned} \bullet (f \circ g)(x) &= f(g(x)) \\ &= f(1+x) \\ &= (1+x)^2 \end{aligned}$$

$$\bullet \text{dom}(f \circ g) = \mathbb{R}$$

$$\begin{aligned} \bullet (g \circ f)(x) &= g(f(x)) \\ &= f(x) + 1 \\ &= x^2 + 1 \end{aligned}$$

$$\bullet \text{dom}(g \circ f) = \mathbb{R}$$

• ESEMPIO

$$\bullet f(x) = \sqrt{x} \quad g(x) = 1+x$$

$$\bullet \text{dom } f = [0, +\infty) \quad \text{dom } g = \mathbb{R}$$

$$\begin{aligned} \cdot (f \circ g)(x) &= f(g(x)) \\ &= f(1+x) \\ &= \sqrt{x+1} \end{aligned}$$

$$\begin{aligned} \cdot \text{dom}(f \circ g) &= \{x \in \mathbb{R} : x+1 \geq 0\} \\ &= [-1, +\infty) \end{aligned}$$

$$\begin{aligned} \cdot \text{dom}(f \circ g) &= \{x \in \mathbb{R} : 1+x \in [0, +\infty)\} \\ &= [-1, +\infty) \end{aligned}$$

b5)

$$\cdot f(x) = x^2 \quad g(x) = \frac{1}{1+x}$$

$$\cdot h(x) = 15x$$

$$\cdot (h \circ g \circ f)(x) = h(g(f(x)))$$

$$= h(g(x^2))$$

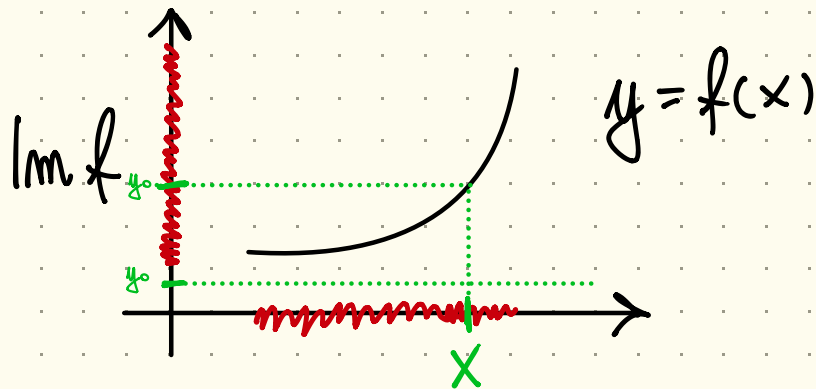
$$= h\left(\frac{1}{1+x^2}\right)$$

$$= 15 \frac{1}{1+x^2} = \frac{15}{1+x^2} \geq 1 \rightarrow \neq 0$$

- $\text{dom}(h \circ g \circ f) = \mathbb{R}$

• FUNZIONE INVERSA

SIA $f: A \rightarrow \mathbb{R}$ (INIETTIVA)



- $\forall y_0 \in \mathbb{R}$ L'EQUAZIONE

$$f(x) = y_0$$

HA AL PIÙ UNA SOLUZIONE $\begin{cases} 1 & y_0 \in \text{Im } f \\ 0 & y_0 \notin \text{Im } f \end{cases}$

- $\forall y_0 \in \text{Im } f$ L'EQUAZIONE

$$f(x) = y_0$$

HA UNA ED UNA SOLA SOLUZIONE

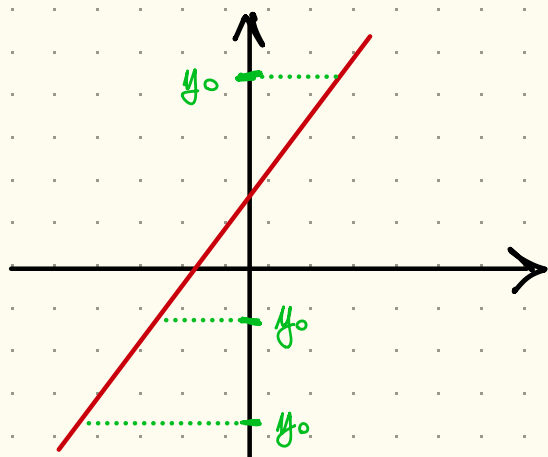
- LA FUNZIONE INVERSA

$$f^{-1}: \text{Im } f \rightarrow \mathbb{R}$$

$$f^{-1}(y_0) = x \text{ TALE CHE } f(x) = y_0$$

es)

• $f(x) = x + 1 \quad f: \mathbb{R} \rightarrow \mathbb{R}$



• f INIETTIVA

• $f'(y) = y - 1$

• $f(y - 1) = y - 1 + 1 = y$

• PROPRIETA'

• $f(f^{-1}(y)) = y \quad \forall y \in \text{Im } f$

• $f^{-1}(f(x)) = x \quad \forall x \in A$

$f(x)^{-1} = 1$

Consideriamo la catena di uguaglianze

$\frac{1}{f(x)} = f(x)^{-1} = f^{-1}(x)$. È corretta? Spiegarne il motivo.

$$\frac{1}{f(x)} = f(x)^{-1} = f^{-1}(x)$$

VERO

$$a^{-1} = \frac{1}{a}$$

INVERSO:
PRODOTTO

$$\frac{1}{a} \cdot a = 1$$

INVERSA:
COMPOSIZIONE